Lecture 36 : Resource Allocation

Introduction

In the intricate world of software project management, successful execution hinges not just on meticulous planning of activities, but crucially, on the intelligent allocation of resources. While traditional project scheduling techniques, such as Critical Path Method (CPM) and Program Evaluation and Review Technique (PERT), excel at sequencing tasks and identifying critical paths, they often overlook a fundamental constraint: the availability of resources. A project plan, no matter how well-structured, remains an ideal blueprint if it doesn't account for the tangible assets and personnel required to bring it to life.

This lecture delves into the critical domain of resource allocation, bridging the gap between theoretical activity plans and practical project execution. We explore how to integrate resource considerations into project scheduling, ensuring that the right resources are available at the right time and in the right quantities. Furthermore, we examine strategies for adapting project plans when resources are scarce, assessing the efficacy of such changes to maintain project viability. The ultimate goal is to produce comprehensive schedules—activity, resource, and cost—that reflect a realistic and achievable path to project completion, transforming abstract plans into actionable strategies.

Core Concepts

Term	Definition	Significance
`Resource`	Any item or person essential for the successful execution of a project, ranging from human capital to physical assets and financial backing.	Fundamental building block of any project; its availability and proper management directly impact project feasibility and success.
`Activity Schedule`	A detailed plan indicating the projected start and completion dates for each individual activity within a project.	Provides a timeline for tasks, crucial for tracking progress and coordinating efforts across the project lifecycle.

Term	Definition	Significance
`Resource Schedule`	A plan that specifies the dates on which each required resource will be needed and the precise level of that requirement.	Ensures resources are available when activities demand them, preventing bottlenecks and optimizing utilization.
`Cost Schedule`	A projection of the cumulative expenditure expected to be incurred through the use of resources over the project's duration.	Essential for financial planning and control, allowing project managers to monitor spending against budget.
`Precedence Network`	A graphical representation of project activities and their dependencies, showing the sequence in which tasks must be performed.	Serves as the basis for activity scheduling and for identifying resource requirements for each task.
`Resource Requirement List`	A comprehensive inventory detailing all resources needed for a project, including their types, quantities, and expected demand levels.	A foundational document for resource allocation, enabling systematic planning and procurement.
`Resource Histogram`	A graphical tool used to visualize resource loading over time, showing the demand for specific resources across the project timeline.	Helps in identifying periods of over-allocation or under-utilization, facilitating resource leveling and optimization.
`Secondary Resource`	A resource, like money, that is primarily used to acquire or facilitate the use of other resources rather than being directly consumed by project activities.	Highlights the indirect but critical role of certain resources in enabling project operations and acquiring primary resources.

Detailed Analysis

1. The Nature and Categories of Resources

Resources are the lifeblood of any project, encompassing anything required for its successful execution. Their nature can vary significantly; some, like a project manager, are needed for the entire duration, while others, such as a programmer for a specific coding task, are required only for particular activities. Effective resource allocation begins with a clear understanding of these distinctions and a systematic categorization of all potential resources.

The lecture identifies seven key categories of resources:

- **Labour**: This category includes all human personnel involved in the project, such as project managers, system analysts, programmers, designers, and testers. These are the members of the development team whose skills and time are directly applied to project tasks.
- **Equipment**: This covers all computing and other physical tools necessary for the project. Examples include servers, workstations, desktops, PCs, laptops, keyboards, printers, scanners, and even accessories like furniture. These are assets that are used over a period rather than consumed.
- **Materials**: Unlike equipment, materials are items that are consumed during the project. This includes items like disks, CDs, and paper, which are used up or become part of the final product.
- **Space**: This refers to the physical working environment required for the project team, such as office space or dedicated work areas where team members can perform their tasks.
- **Services**: Some projects necessitate specialized external services. For instance, developing a wide-area distributed system might require telecommunication services, or a project might need postal or courier services. These are external supports that facilitate project activities.
- **Time**: Considered an important resource, time here refers to "elapsed time." The lecture notes that elapsed time can sometimes be reduced by increasing other resources, such as adding more staff to meet tight deadlines.
- **Money**: While crucial, money is categorized as a secondary resource. Its primary function is to acquire other resources. It is similar to other resources in that it comes at a cost, often in the form of interest charges if borrowed.

2. Resource Allocation Process Overview

The process of resource allocation involves several critical steps to ensure that resources are effectively matched to project activities:

1. **Identify Resources Needed for Each Activity**: The first step is to meticulously determine what specific resources are required for every single activity within the project.
2. **Create a Resource Requirement List**: Based on the identified needs, a comprehensive list is compiled, detailing all resources, their types, and the expected level of demand.
3. **Identify Resource Types**: This involves categorizing resources, recognizing that individuals within a group might be interchangeable (e.g., any "VB programmer" rather than a specific individual).
4. **Allocate Resource Types to Activities**: Once types are identified, they are assigned to the relevant activities.
5. **Examine the Resource Histogram**: A resource histogram is then used to visualize resource loading over time, allowing project managers to assess if resources are uniformly allocated and to identify potential over-allocation or under-utilization.

3. Identifying Resource Requirements - Hardware

Identifying hardware requirements is a systematic process involving several steps:

1. **List All Hardware Needs**: Begin by creating a comprehensive list of every piece of hardware the project will require. This could include desktops, servers, backup media, keyboards, switches, networking ports and cables, display units, printers, plotters, and touchscreens.
2. **Specify Function for Each Piece**: For each listed hardware item, clearly define its purpose and what it is expected to do within the project.
3. **Specify Needs for Proper Functioning**: Determine what each piece of hardware requires to function correctly. This might include specific software tools, power requirements, or interconnection cables.
4. **Assign Hardware to Users**: Identify which individuals or roles need specific hardware to perform their tasks. It's crucial not to over-provision; provide only what is necessary for each person's role.
5. **Specify When Hardware is Needed**: Determine the timeline for hardware acquisition. Identifying needs in advance can facilitate procurement and prevent delays.

4. Identifying Resource Requirements - Software

Software resources are equally critical but present unique challenges due to their dynamic nature. Basic software like operating systems and compilers, if incorrectly installed or outdated, can cause significant project delays. Mismatched versions, service packs, or library releases can also lead to severe problems. Experienced project managers understand that random software upgrades during a project can severely impact team productivity.

Therefore, a planned approach to software management is essential:

1. **List Software Requirements**: The project manager must compile a detailed list of all software products the team will need. This includes:
2. Operating systems
3. Compilers
4. Configuration management systems
5. Email systems
6. Supporting software (e.g., FTP, web browsers)
7. Libraries (e.g., dynamic link libraries, standard template libraries, GUI libraries)
8. Office software (e.g., word processing, spreadsheets, presentation tools)
9. Software tools (e.g., CASE tools, design tools like RSA, testing tools like JUnit, project management tools like Gantt chart tools).
   This detailed list helps minimize inconsistencies, plan upgrades, and manage licensing.
10. **Specify Software Versions**: For each software product, the exact version required by the team must be specified. This also includes anticipating when changes or new versions might become available.
11. **Identify Upgrades and Service Packs**: Since software is not static, upgrades and enhancements are inevitable. The project manager needs to know which software pieces will require upgrades and when these will occur. Key questions to answer include:
12. How does the project's length relate to potential upgrades, service packs, or new versions?
13. Will the team need to upgrade?
14. When will upgrades be available?
15. **Identify Who Upgrades Which Software**: Clearly define who is responsible for installing and upgrading each piece of software. This could

be a system administrator or a specific project team member. Everyone involved must agree on these responsibilities.

16. **Specify When to Upgrade Which Software**: Timing upgrades is critical. Upgrading just before a major milestone is highly risky, as unforeseen problems could jeopardize the milestone. Upgrading immediately after a milestone allows time for troubleshooting but might invalidate prior testing. The best approach is to plan upgrades when the team and project can best handle unexpected problems, anticipating that issues will occur and allocating time to address them. This proactive planning ensures that if problems arise, there's a buffer, and if not, the team gains extra time.

5. Identifying Resource Requirements - Support Staff

Beyond the core development team, support staff play a vital role. Identifying and managing their involvement is crucial:

1. **Identify Support Needed from Each Group**: Determine what kind of support is required from various internal or external groups.
2. **Specify When Support Will Be Required**: Pinpoint the exact times or project phases when support will be needed to ensure continuous progress without delays.
3. **Specify How Support Occurs**: Define the mode of support delivery (e.g., physical presence, phone, email).
4. **Gain Commitment**: Secure formal commitment for the required support, whether through verbal agreement, a contract, or a commitment letter.
5. **Maintain Good Relationships**: Foster positive relationships with support staff, as this is essential for smooth project execution.

6. Preparing a Resource Requirement List

The culmination of identifying all resource needs is the creation of a comprehensive resource requirement list. This list is the first step in producing a resource allocation plan. It details all resources, their expected demand levels, and their duration.

This list is typically prepared by analyzing each activity within a precedence network (or activity network). For example, a project manager is a resource required for the entire project duration, not just specific activities. Similarly, workstations might be needed across multiple stages. The list would include:

- **Stage/Activity**: The specific project phase or task.
- **Resource**: The type of resource (e.g., Project Manager, Senior Analyst, Workstation).

- **Days**: The number of days the resource is required (e.g., 104 F/T for a project manager, 34 F/T for a senior analyst).
- **Quantity**: The number of units of that resource (e.g., 34 workstations).
- **Notes**: Any additional remarks, such as software availability checks or specific usage conditions.

By systematically populating this list from the precedence network, project managers gain a clear overview of all resource demands, which then informs the actual allocation process.

Key Takeaways

- [ ] Resource allocation is crucial for bridging the gap between theoretical project plans and practical execution.
- [ ] Project scheduling techniques alone are insufficient without considering resource availability.
- [ ] The final output of resource allocation includes activity, resource, and cost schedules.
- [ ] Resources are categorized into labour, equipment, materials, space, services, time, and money, each with distinct characteristics.
- [ ] Money is considered a secondary resource, primarily used to acquire other resources.
- [ ] Identifying hardware requirements involves listing needs, specifying functions, determining dependencies, assigning to users, and timing acquisition.
- [ ] Software management requires careful planning for versions, upgrades, and responsibilities to avoid project delays and productivity loss.
- [ ] Upgrading software should be planned strategically, avoiding periods just before major milestones and anticipating potential problems.
- [ ] Support staff needs to be identified, their availability specified, commitment secured, and good relationships maintained.
- [ ] A comprehensive resource requirement list, derived from the precedence network, is the foundational document for resource allocation.

Conclusion

Effective resource allocation is not merely an administrative task; it is a strategic imperative for successful software project management. By systematically identifying, categorizing, and planning for all necessary resources—from human capital and equipment to software and support services—project managers can

transform abstract project schedules into realistic and achievable roadmaps. The detailed steps for identifying hardware, software, and support staff, coupled with the creation of a robust resource requirement list, provide a structured approach to managing the complex interplay of project activities and their resource demands.

This lecture has laid the groundwork for understanding how to integrate resource considerations into every phase of project planning. By anticipating challenges related to software upgrades, judiciously assigning hardware, and securing necessary support, project teams can mitigate risks, optimize efficiency, and ultimately deliver projects on time and within budget. The principles discussed here are fundamental to navigating the practical realities of project execution, ensuring that resources are not just available, but are deployed with maximum impact.